

Adaptive Wildfire Governance Through Reputation-Weighted Response Networks

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Abstract

This paper proposes an adaptive governance framework for wildfire response based on reputation-weighted coordination among distributed response actors. The model combines reputation scoring, resource allocation, and a circulation-inspired incentive mechanism to improve the timeliness, resilience, and accountability of wildfire governance in high-risk landscapes. A lightweight simulation of representative fire-management agencies demonstrates how reputation-weighted decision influence can support faster consensus on resource deployment and more transparent post-event evaluation. The paper contributes a reproducible conceptual prototype that bridges wildfire governance, trust-weighted collaboration, and adaptive response design.